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Benefits of oxygen enrichment in sponge iron rotary kiln emerging from data-driven modelling

N. Rajesh*[1,2], A. Nandi[3] and S. K. Pabi[1]

Optimal oxygen enrichment conditions for sponge iron rotary kiln have been successfully explored on an industrial scale using a data-driven model. A multi-objective optimisation by genetic algorithm (MOGA) is employed to find the favourable conditions. The objective function for MOGA is derived from neural networks using pre-processed operational data. From industrial experimentations guided by the optimum conditions predicted by the present model, it emerged that when the coal fines injection is maintained at 1.75 tph and the oxygen enrichment is 8 $Nm^3\,t^{-1}$ of sponge iron, a reduction in the specific air requirement from 2609 to 2150 $Nm^3\,t^{-1}$ was obtained, while the end-zone bed temperature remained under control at 1132°C. These conditions resulted in a reduction of specific coal consumption by 6%, an enhancement in the sponge iron production by 6% and an increase in the rotary kiln campaign life from 50 to 100 days.

Keywords: Sponge iron rotary kiln, Oxygen enrichment, Data-driven model, Optimisation, Genetic algorithm, Artificial neural network

Introduction

Sponge iron manufacturers using rotary kilns are exploring new avenues to improve the performance of the process because of the limited supply of high-quality raw materials, stringent environmental norms and increased overheads. Focus is mainly on the increase in productivity, reduction in coal consumption and enhancement of kiln campaign life.[1] Conventional sponge iron production in a rotary kiln[2,3] uses raw materials such as iron ore or iron pellets, coal and fluxes (dolomite/limestone). The use of oxygen-enriched air in an industrial sponge iron rotary kiln has not been explored so far to the best of our knowledge, but the benefits of oxygen are well established[4] in other iron ore reduction processes such as the blast furnace. Oxygen enrichment of air in a rotary kiln potentially can lead to oxidation of iron and temperature overshoot in the kiln, which can lead to the early formation of accretions and hence a reduction in the kiln campaign life. These are detrimental to a widespread use of oxygen in the sponge iron rotary kiln. Controlled oxygen enrichment may, however, eliminate all these problems. Oxygen enrichment can lead to a reduction in the nitrogen load, which would eventually help reduce fuel consumption. Thus, it would be interesting to find the optimum oxygen requirement to derive the beneficial effects of oxygen enrichment in the sponge iron rotary kilns.

A number of studies reported in the literature[5–11] have attempted to model the rotary kiln; but the models incorporating the use of oxygen-enriched air in a sponge iron rotary kiln are yet to be reported to the best of our knowledge. Therefore, the present study explores the conditions required to drive the beneficial use of oxygen in the sponge iron rotary kiln through a modelling exercise. The model has been developed for application under industrial conditions. The operational details of sponge iron rotary kilns, methodology adopted for the development of the model of oxygen enrichment, results of the proposed model, interpretation of the effect of various input variables on the output of the model and finally the implementation of the model in actual practice in a typical plant are discussed in the following sections.

Operational details of sponge iron rotary kiln with oxygen enrichment

A schematic diagram of the sponge iron rotary kiln is displayed in Fig. 1. Here, the solid raw materials and gases move in a counter-current direction. Even though iron-bearing material is charged from the feed end, a part of the total coal required for the process is typically injected from the discharge end to supplement the heat required for reduction at the end-zones of the kiln. In general, air is introduced into the freeboard to burn the carbon monoxide evolved from the bed to provide the heat required for the process, in particular for the endothermic Boudouard reaction.[11,12] In a controlled reducing atmosphere, the counter-current movement of gases over the moving bed of solids allows for proper reaction and heat transfer between the solid and the gases, so reducing the iron bearing material. A suitable balance of conditions should be maintained to minimise eutectic melting.

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1 **Schematic diagram of a sponge iron rotary kiln**

The quality of sponge iron is determined by the reduced iron and is termed as 'iron metallic' (%Fe(M)), which in turn depends mainly on the quality of input raw materials, operational parameters, thermal and air profiles and retention time available inside the kiln.[10]

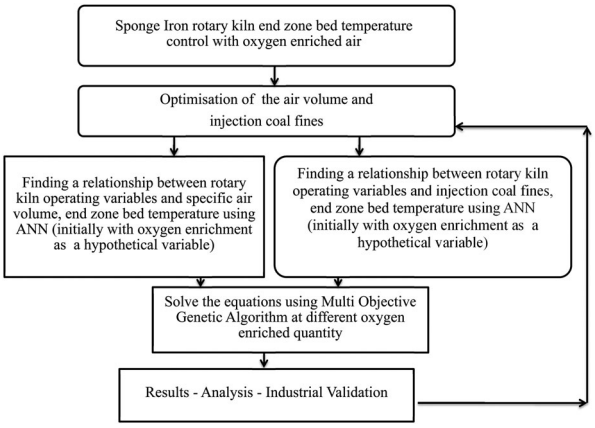
In sponge iron rotary kilns operating under normal conditions with or without oxygen enrichment, with continuous supply of raw material from a single source and without any equipment breakdown, accretion[11] is the major cause for the end of the campaign. A sensitive balance in the kiln-operating condition has to be maintained for obtaining a quality product at relatively low temperature, wherein no eutectic melting occurs inside the kiln.[13] The temperature of the material in the kiln is known as bed temperature, whereas the freeboard temperature is known as gas temperature.[14] Controlling the rotary kiln bed temperature at the end-zone (BTEZ) is one of the key parameters for prolonged campaigns and improving productivity. An increase in the partial pressure of reducing gases increases the rate of reduction. For this a controlled amount of oxygen can be injected into the kiln along with air containing the normal ratio of nitrogen and oxygen. It is known that some nitrogen is required as a dilutant for controlling the oxidation rate of iron, but at a cost of increased carbon consumption.

Oxygen enrichment should be able to bring down the excess air requirement[15,16] and the amount of coal consumed in the process of heating the excess air. Increased oxygen availability inside the rotary kiln promotes proper combustion of coal, which in turn raises the temperature. This helps in accommodating the excess iron feed material for the same coal consumption and thus, it may be possible to improve production per tonne of sponge iron at reduced coal consumption. However, the temperature of the kiln would increase by introducing oxygen, and hence there is a chance of increased accretions inside the kiln. Therefore, controlling the temperature would be a key to the success of using oxygen-enriched air. The temperature rise in the end-zone of the kiln can be restricted by a coolant, i.e., by the coal fines injection (CFI). The amount of CFI and the air quantity to be supplied need to be determined for a specific quantity of oxygen, and this eventually becomes a multi-objective optimisation problem.

Model design

Artificial neural network used

The scheme of the multi-objective optimisation program to determine the favourable conditions for oxygen enrichment is illustrated in Fig. 2. In the first model the BTEZ and the quantity of specific air have been optimised, whereas in the second model the BTEZ and CFI are optimised. The objective function for this optimisation was derived from the artificial neural network (ANN) model. Based on the plant data for air-based operation without oxygen enrichment, the number of input variables for each ANN model is selected and input–output fitting models for specific-air requirement, CFI and BTEZ of the rotary kiln have been designed. The best suited neural network architecture is then taken as the objective functions and using a multi-objective genetic algorithm (MOGA) optimisation procedure, the process parameter values are predicted, which will be nearest to the optimal value for the desired BTEZ of the rotary kiln.



2 **Model development strategy for controlling BTEZ in the sponge iron rotary kiln**

Information on the sponge iron rotary kiln process from a selected plant (DRI CORP) has been used as the input and output variables in the present model. DRI CORP is sourcing its major raw materials (iron ore and coal) from limited suppliers with the intention to keep quality parameters (for coal, reactivity, volatile matter, etc., and for iron ore, size range, reducibility, etc.) within a narrow range. The operational details of DRI CORP are given in Table 8. For the first neural network for predicting specific-air volume requirement, a list of six variables has been identified (Table 1). Similarly, four variables, as listed in Table 2, are identified for the second neural network for CFI determination. Final inputs are selected based on the correlation analysis[17] and those exhibiting low correlation are removed on the basis of sensitivity analysis. In this process, coal moisture, reactivity, volatile matter, iron ore reducibility, kiln rotation, i.e.,

Table 1 Input and output variables for optimising BTEZ and quantity of specific air

Inputs	Outputs
1. Iron ore feed rate/tph	1. BTEZ/°C
2. Feed coal/tph	2. Quantity of specific air/$Nm^3\ t^{-1}$ of iron ore feed
3. Quantity of coarse coal injection/tph	
4. CFI/tph	
5. C/Fe ratio	
6. Enriched oxygen quantity/$Nm^3\ t^{-1}$	

Table 2 Input and output variables for optimising BTEZ and quantity of CFI

Inputs	Outputs
1. Iron ore feed rate/tph 2. Quantity of specific air/Nm^3 t^{-1} of iron feed 3. C/Fe ratio 4. Enriched oxygen quantity/Nm^3 t^{-1}	1. BTEZ/°C 2. Quantity of CFI/ tph

number of revolutions per minute, etc., did not became the input and output variables for the models.

In both the models BTEZ is chosen as one of the output variables. Descriptive statistics of all the input and output variables used in both the models are shown in Table 3 and they are: iron ore feed rate to the rotary kiln expressed in tonnes per hour (tph), feed coal to the rotary kiln expressed as tonnes per hour (tph), quantity of coarse coal injected from the discharge end having a size fraction of 4–8 mm, expressed in tph, coal fines injected at the discharge end of size 1–4 mm, expressed in tph, specific air is the amount of air supplied given as normal cubic metre per tonne of iron ore feed (Nm^3 t^{-1}) and C/Fe ratio, i.e., the ratio between total carbon present in the feed and total iron in the feed. Enriched oxygen quantity is the additional oxygen supplied expressed as normal cubic metre per tonne of sponge iron quantity (Nm^3 t^{-1}). Bed temperature at the end-zone is the temperature (°C) prevailing at the end-zone of the rotary kiln measured by a thermocouple, which is fixed on the shell of the kiln.

Root mean square error (RMSE) and correlation coefficient (R^2) are used as the performance metrics for the present models (equations (1) and (2)), as follows:

$$\mathrm{RMSE} = \frac{1}{n}\sqrt{\sum_{i=1}^{n} |pi - oi|^2} \tag{1}$$

$$R^2 = 1 - \left\{\sum |pi - oi|^2 / \left(\sum |pi - oi|^2 + \sum |pi - oi\ bar|^2\right)\right\} \tag{2}$$

Here pi is the predicted value produced by the model, oi is the observed value, $oi\ bar$ is the mean of the observed value and n represents number of test points.

Training of neural network

Specific air and BTEZ model involving the selected parameters are expressed by equation (3), as follows:

$$y = f(x1, x2, x3, x4, x6, x7) \tag{3}$$

The CFI and BTEZ model involving the selected parameters are expressed by equation (4):

$$y = f(x1, x5, x6, x7) \tag{4}$$

For both the models with the given set of input and output conditions, simulations were conducted to determine the optimum number of neurons in the hidden layers, number of hidden layers, learning rate and the momentum factor using the software package MATLAB.

Results and discussion

Objective function determination

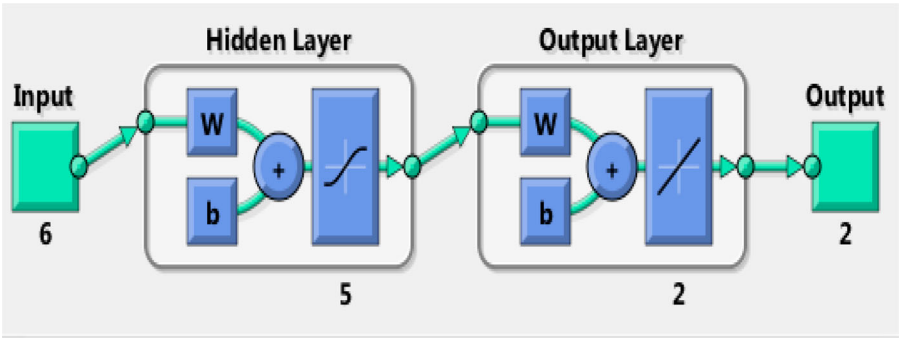
Objective function determination for bed temperature at the end-zone, specific-air consumption and coal fines injection using feed forward neural network

A typical ANN architecture for prediction of BTEZ and specific-air consumption is shown in Fig. 3. It is a three-layer supervised back-propagation network. There are six input neurons corresponding to the six input variables selected for the model, one hidden layer consisting of five neurons and two output neurons. These neurons are interconnected to each neuron of the previous and the next layers. The connection between two neurons has been parameterised by a weight and a bias. Sigmoid and linear activation functions were selected for the hidden and output layers, respectively. The network architecture shown in Fig. 3 has been trained in the standard manner with 150 sets of training data for the sponge iron rotary kiln through ≤5000 iterations. Further, the convergence has been tested and when the change in error between consecutive iterations becomes less than 10^{-2}, the training has been stopped. In the training phase, it has been ensured that the final error in the validation phase converges to a value less than the final value of error. For the training model, it became apparent that the error in

Table 3 Statistics of the process parameters

	Parameter	Unit	Minimum	Maximum	Mean	Standard deviation
$x1$	Iron ore feed	tph	6	7	6.4	0.2045
$x2$	Feed coal	tph	1.85	2.2	1.93	0.03
$x3$	Quantity of coarse coal injection	tph	0	0.63	0.35	0.2
$x4$	Quantity of CFI	tph	1.71	2.5	2.01	0.23
$x5$	Specific air	Nm^3 t^{-1}	2100	2800	2409	177
$x6$	C/Fe ratio		0.42	0.52	0.48	0.15
$x7$	Enriched oxygen quantity	Nm^3 t^{-1}	0	30	19	3.5
$x8$	BTEZ	°C	1060	1160	1125	13.8

Iron ore feed rate (tph), the amount of iron ore fed to the rotary kiln tonnes per hour; feed coal (tph), the amount of feed coal fed to the rotary kiln tonnes per hour; quantity of coarse coal injection (tph), coal that is injected from the discharge end having a size fraction of 4–8 mm; CFI (tph), coal that is injected from the discharge end having a size fraction of 1–4 mm; specific air (Nm^3 t^{-1} of iron ore feed), the amount of air supplied normal cubic metre per tonne of iron ore feed; C/Fe ratio, the ratio between total carbon present in the feed and total iron in the feed; enriched oxygen quantity (Nm^3 t^{-1}), the amount of additional oxygen supplied normal cubic metre per tonne of sponge iron quantity; BTEZ (°C), the temperature prevailing in the end-zone of the rotary kiln measured by a thermocouple that is fixed on the shell of the kiln.



3 **Feed forward neural network for controlling BTEZ and specific-air consumption in the sponge iron rotary kiln (from MATLAB, 2010)**

Table 4 The best network topologies for controlling BTEZ and specific-air consumption in the sponge iron rotary kiln

Network topology	Training error	Validation error	Test error
6–10–6–2	0.068	0.074	0.068
6–5–2	0.049	0.060	0.057

Table 5 The best network topologies for controlling BTEZ and CFI in the sponge iron rotary kiln

Network topology	Training error	Selection error	Test error
4–10–6–2	0.076	0.077	0.065
4–7–2	0.038	0.055	0.057

the output decreased sharply with the number of neurons and then reached a plateau. Further increase in the number of neurons did not result in any substantial improvement in the network performance. In case of number of hidden layers, the use of single layer gave distinctly superior performance as compared to multiple layers for the learning rate of 0.05, and momentum factor of 0.2. Results depicting the performance of the model are shown in Table 4. For a single hidden layer network, best performance of the model has been obtained with five neurons for specific air and BTEZ model with the sigmoid hidden activation functions in the hidden layer and a linear output activation function for the output neurons (Table 4).

The methodology described above has been repeated to find the relation between CFI and BTEZ with the input operating variables of the sponge iron rotary kiln. The results depicting the performance of the prediction model are shown in Table 5. In reference to a single hidden layer network, best performance for the prediction model has been obtained with seven neurons for CFI and BTEZ with the sigmoid hidden activation function in the first hidden layer and a linear output activation function for the output neurons.

Optimisation

Optimisation of bed temperature at the end-zone and specific-air volume using the selected feed forward neural network as objective function

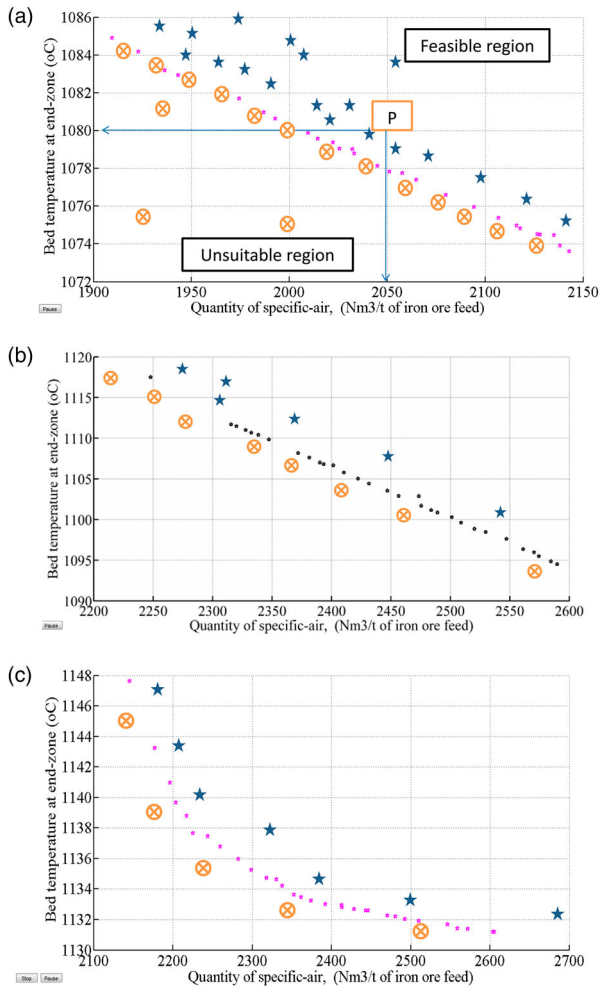
Multi-objective genetic algorithm[18] has been used for optimisation of the sponge iron rotary kiln process parameters to obtain the minimum BTEZ and specific-air quantity. Trained feed forward neural network has been used as the objective function for the optimisation model. The code has been developed on MATLAB[19] platform. In order to optimise the specific-air requirement for the desired BTEZ, the multi-objective problem can be expressed as a function of control variables. When considering the operation conditions at the selected plant, the iron ore feed rate $x1$ has been constrained between 6 and 7 tph, feed coal quantity $x2$ has been constrained between 1.8 and 2.2tph, injection coarse coal quantity $x3$ has been constrained between 0 and 0.63tph, CFI quantity $x4$ has been constrained between 1.71 and 2.5tph, C/Fe ratio $x6$ has been constrained between 0.42 and 0.52 and oxygen injection rate in air $x7$ has been constrained between 1 and 30 $Nm^3\,t^{-1}$. The multi-objective optimisation problem can then be presented as:

$$\begin{aligned} &\min f(x1, x2, x3, x4, x6, x7) \\ &\text{Subject to the constraints} \\ &6 \le x1 \le 7, \\ &1.8 \le x2 \le 2.2, \\ &0 \le x3 \le 0.63, \\ &1.71 \le x4 \le 2.5, \\ &0.42 \le x6 \le 0.52 \text{ and} \\ &1 \le x7 \le 30 \end{aligned} \tag{5}$$

Here f is the function in equation (3), relating input parameters of specific air and BTEZ. The descriptions of the six input variables of model (5) are shown in Table 1. The test set was used to solve model (5) with the MOGA available in MATLAB.[19]

In multi-objective optimisation, values of objective functions create a multi-dimensional space called objective space. Each decision variable on variable space corresponds to a point in objective space. A solution that does not satisfy all the constraints and variable bounds is referred to as an infeasible solution, whereas which satisfies all the conditions is known as a feasible solution. The Pareto optimal is the best known (optimal) solution with respect to all objectives and cannot improve any objective without worsening the another objective. The set of Pareto optimal outcomes is often called the Pareto front.

A local Pareto front is identified for multiple objective functions using the genetic algorithm. The Pareto plot displays two competing objectives, specific air and



4 Pareto front GA corresponding to various oxygen input ranges *a* oxygen input range of 1–10 $Nm^3 t^{-1}$; *b* oxygen input range of 11–20 $Nm^3 t^{-1}$; *c* oxygen input range of 21–30 $Nm^3 t^{-1}$. Dots, ... best solution; asterisk, ★ feasible region; crossed circle, ⊗ non-feasible region

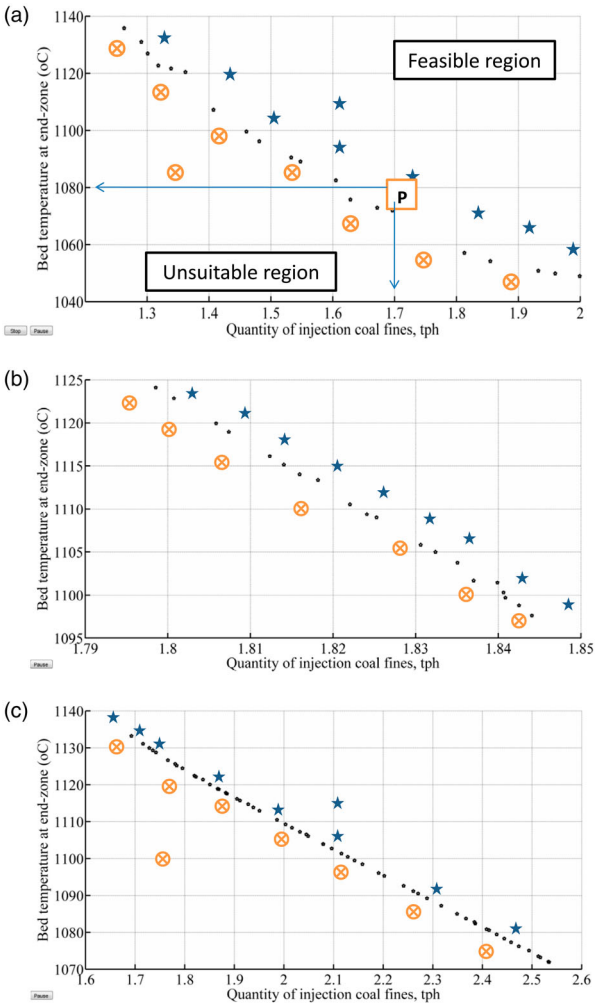
BTEZ. The MOGA works on a population using a set of operators that are applied to the population. The initial population has been generated randomly by default. The next generation of the population has been computed using the non-dominated rank and a distance measure of the individuals in the current generation. When maximum number of iterations has been reached, then the solver stops. The initial parameters of the MOGA are as follows: the population size is 65 and the maximum number of iterations is set at 100. Three different Pareto fronts are generated with varying oxygen input ranging from 1 to 10, 11 to 20 and 21 to 30 $Nm^3 t^{-1}$ and are displayed in Fig. 4. Here Fig. 4*a* shows the Pareto front corresponding to an oxygen input range of 1–10 $Nm^3 t^{-1}$ of iron ore, the second Pareto front shown in Fig. 4*b* corresponds to an oxygen input range of 11–20 $Nm^3 t^{-1}$ of iron ore and the third Pareto front shown in Fig. 4*c* corresponds to an oxygen input range of 21–30 $Nm^3 t^{-1}$ of iron ore. Different Pareto frontiers are generated to explore the effects of increased oxygen input on the target objectives. The regions marked asterisks are in the feasible region, whereas crossed circles delineate to the non-feasible region. The best solutions are on the optimal front represented by 'dots' in Fig. 4*a*. Same is the case with other Pareto fronts shown in Fig. 4*b* and *c*. With reference to the Pareto front generated, if the BTEZ requirement is, say, 1080°C, for oxygen enrichment between 1 and 10 $Nm^3 t^{-1}$, the specific-air requirement would be 2050 $Nm^3 t^{-1}$ (point P in Fig. 4*a*). On the other hand, for oxygen enrichment level between 10 and 30 $Nm^3 t^{-1}$, to maintain the BTEZ of 1080°C the specific-air requirement is exceeding 2600 $Nm^3 t^{-1}$. The value shows that with increase in oxygen consumption end-zone temperature (BTEZ) of the rotary kiln has been increasing, and then in order to bring down that temperature, the specific-air volume needs to be increased, which implies that the purpose of oxygen enrichment would fail at the higher level of oxygen enrichment. On the basis of these results, it appears that a specific oxygen enrichment of <10 $Nm^3 t^{-1}$ of sponge iron may yield optimum results in the sponge iron rotary kiln.

Optimisation of bed temperature at the end-zone and coal fines injection using the selected feed forward neural network as objective function

Similarly, in order to optimise the CFI to get the desired BTEZ, the multi-objective problem can be expressed as a function of control variables as in (6):

$$\begin{aligned} &\min f(x1, x5, x6, x7) \\ &\text{Subject to the constraints:} \\ &6 \le x1 \le 7, \\ &2100 \le x5 \le 2800, \\ &0.42 \le x6 \le 0.52 \text{ and} \\ &1 \le x7 \le 30 \end{aligned} \tag{6}$$

Here f is the function in equation (4). The descriptions of the four input variables of equation (6) are shown in Table 2. For controlling the temperature, one possible way is CFI. The results of the CFI quantity and BTEZ optimisation study are presented in Fig. 5. Here Fig. 5*a* shows the Pareto front corresponding to specific oxygen input range of 1–10 $Nm^3 t^{-1}$on of sponge iron, the second Pareto front shown in Fig. 5*b* corresponds to a specific oxygen input range of 11–20 $Nm^3 t^{-1}$on of sponge iron and the third Pareto front shown in Fig. 5*c* corresponds to a specific oxygen input range of 21–30 $Nm^3 t^{-1}$ of sponge iron. Different Pareto frontiers are generated to explore the benefits of increased oxygen input on the target objectives. The values show that with increase in oxygen consumption the BTEZ of the rotary kiln has been increasing, (y-axis in Fig. 5) and to curtail the increase in BTEZ, the CFI quantity needs to be increased. The Pareto front generated for the 1080°C BTEZ (point P in Fig. 5*a*) evidences that there is a need to inject additional quantity of coal fines into the rotary kiln. More than 2.5 tph CFI, amounting more than 60% of the total coal feed given to the rotary kiln, is required when the specific oxygen quantity has been between 10 and 30 $Nm^3 t^{-1}$. This condition may provoke coal starvation[20] in other areas of the rotary kiln and may not be suitable for kiln operation. Therefore, in order to maintain a low BTEZ (say, BTEZ = 1080°C), a specific oxygen enrichment of <10 $Nm^3 t^{-1}$ of sponge iron can be beneficial in sponge iron rotary kiln. Several input data sets are generated by the optimisation program for a minimum BTEZ, specific-air quantity requirement and CFI. The Pareto front for all these data sets denotes the best possible



5 Pareto front GA corresponding to various oxygen input ranges for CFI and BTEZ; *a* oxygen input range of 1–10 Nm^3 t^{-1}; *b* oxygen input range of 11–20 Nm^3 t^{-1}; *c* oxygen input range of 21–30 Nm^3 t^{-1}. Dots, ... best solution; asterisk, ★ feasible region; crossed circle, ⊗ non-feasible region

tradeoffs between the objectives. While considering the plant-specific operational experience, the correct variable suitable for the specific situations of the kiln has to be chosen first. This is also regarded as a standard practice in the multi-objective optimisation literature.[8] For the objective of achieving a minimum BTEZ of 1080°C and specific-air consumption of 2050 Nm^3 t^{-1}, the set of inputs corresponding to these values needs to be selected on the basis of operational experience of the plant personnel. Table 6 presents such typical data set. A comparison of the average values of actual plant data and the optimised set has been presented in Table 7. The optimised data in Table 7 reveal that there has been a possibility to operate at minimum end-zone temperature and specific-air quantity at low oxygen enrichment of 1–10 Nm^3 t^{-1}. From the Pareto plots drawn further at different oxygen enrichment levels of 1–4, 5–7 and 8–10 Nm^3 t^{-1}, it is shown that a specific oxygen enrichment of 8–10 Nm^3 t^{-1} of sponge iron may give optimum results in the sponge iron rotary kiln.

Validation against industrial data

The present model has been validated and the following details are reported at one 100-tpd sponge iron rotary kiln in DRI CORP. This pseudo name has been used for data confidentiality. The details of the industrial data are reported in Table 8. In an air-based operation the operational parameters are collected and it was found that 2600 Nm^3 t^{-1} specific-air supply is needed for getting the desired product quality of Fe(M) 75%. A daily production of 116 tonnes is reported with a specific coal consumption of 0.92 t per tonne of sponge iron production. On an average, a campaign life of 50 days is achieved previously without oxygen enrichment. The air profile and the temperature profile for getting the desired sponge iron quality are noted. Oxygen is introduced into the kiln from a central blower. By controlling the mechanical dampers, air volume is reduced from each of the shell air fan and central blower, and the temperature profile is altered to maintain same temperature as that for air without oxygen enrichment. Coal fines injection is also adjusted accordingly. Minimal variations are done in the kiln rotation and kiln pressure throughout the campaign.

Table 8 presents the data set with air base and oxygen-enriched condition. The introduction of oxygen increased BTEZ as expected and the heat transfer, which took place inside the kiln, raised the temperature across the length of the kiln. Keeping the total amount of coal supplied to the kiln as that of base condition, the feed rate of the iron ore increased gradually. It was observed that, on an average, the rotary kiln accepted 6% more iron ore feed during the oxygen enrichment period compared to the base line for the same without oxygen enrichment. A campaign life of 100 days has been achieved under this oxygen enrichment condition. Table 8 summarises the current (base) plant condition vs. the oxygen enrichment conditions. From this, it can be concluded that, by controlling the extent of oxygen enrichment of air, regulating the BTEZ by adjusting the specific air and CFI in the sponge iron rotary kiln, the campaign life is doubled as compared

Table 6 Input and output parameters corresponding to 1–10 Nm^3 t^{-1} oxygen

Input	Set 1	Set 2	Set 3	Set 4	Average value
Iron ore feed rate/tph	6.8	6.8	6.9	7.2	6.9
Feed coal/tph	1.95	1.75	1.84	1.9	1.86
Injection coal coarse/tph	0.40	0.65	0.55	0.60	0.55
Injection fine coal feed/tph	1.65	1.75	1.72	1.9	1.75
C/Fe ratio	0.48	0.48	0.47	0.48	0.47
Enriched oxygen quantity/Nm^3 t^{-1}	8	9	8	9	8.5
BTEZ/°C	1078	1075	1088	1082	1080
Specific-air volume/Nm^3 t^{-1}	2060	2030	2055	2040	2050

Table 7 Model development, current plant practice vs. optimised results by genetic algorithm

Input	Average value	Optimised value	Remarks
Iron ore feed/tph	6.4	6.9	Increased iron ore feed
Feed coal/tph	1.93	1.85	Lower feed coal
Quantity of coarse coal injection/tph	0.35	0.55	Increased coarse coal
CFI/tph	2.01	1.75	Lower fine coal
Specific air/$Nm^3\ t^{-1}$	2409	2050	Lower specific air
BTEZ/°C	1080	1082	Remained same
Enriched oxygen quantity/$Nm^3\ t^{-1}$	0	8.5	Oxygen-enriched air

Table 8 DRI CORP operational details, current plant practice (base case) vs. oxygen-enriched case

Input	Base case (air-based operation)	Oxygen-enriched case	% change during oxygen-enriched vs. base case	Remarks
Iron ore feed/tph	7.0	7.42	6%	Increased iron feed
Feed coal/tph	1.93	1.85	–6%	Lower feed coal
Quantity of coarse coal injection/tph	0.35	0.55		Increased coarse coal
CFI/tph	2.01	1.75		Lower fine coal
Specific air/$Nm^3\ t^{-1}$	2600	2150		Lower specific air
BTEZ/°C	1130	1132		Remained same
Enriched oxygen quantity/$Nm^3\ t^{-1}$	0	8		Optimised enriched oxygen quantity
Sponge iron product quality/% Fe(M)	75.4	75.2		Remained same
Campaign life/no. of days	50	100	100	Increased campaign life

to the previous campaign with air-based operation. A 6% improvement in the production and 6% decrease in coal consumption were also observed. At oxygen enrichment of 8 $Nm^3\ t^{-1}$ of sponge iron, the air volume requirement was at 2150 $Nm^3\ t^{-1}$ of sponge production and the BTEZ was maintained at 1132°C. These two values were very close to the optimised model values. It is, therefore, apparent that the model developed here is valid in the industrial domain.

Conclusions

The concept of oxygen enrichment of air in the sponge iron rotary kiln has been explored on industrial scale through artificial neural network and genetic algorithm. The model identifies that, for a specific amount of oxygen enrichment, the quantity of specific air and coal fines injection are controllers of the bed temperature at the end-zone. With increase in oxygen consumption, the bed temperature at the end-zone of the rotary kiln increases and to curtail that, a specific-air volume and coal fines injection quantity has to be increased. Thus, at higher level of oxygen enrichment the benefits of oxygen enrichment in the rotary kiln are minimal. The level of optimal oxygen enrichment for a plant can be evolved by such modelling. During the industrial validation of the present model, oxygen is introduced into the rotary kiln at 8 $Nm^3\ t^{-1}$ of sponge iron. Depending on the level of oxygen introduced in the range of 8–10 $Nm^3\ t^{-1}$, air volume is reduced from 2600 to 2150 $Nm^3\ t^{-1}$. Coal fines injection is maintained at 1.75tph. These conditions kept the bed temperature at the end-zone under control at 1132°C, and resulted in a reduction of specific coal consumption by 6%, with increase in sponge iron production by 6% and increase in the campaign life from 50 to 100 days.

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